

Anchor paper worksheet (1 of 2)

1 Pick ONE anchor paper

- Related to your group's research topic: a biomedical problem plus machine learning
 - Peer-reviewed, last 5 years; cite the published version, not the preprint
- Check: paper criteria
 - ML + biomedical, last 5yr, published, not review
 - Not a review
 - Not from [excluded venues]: [one-line why]
 - Search PubMed, Scholar, WoS or the LLM helper

2 3 Formulate the ML problem, find two more papers

Ignore the model: use the abstract and Fig. 1. Papers 2 and 3 need the same input and output; disease, data source and model may differ.

	Anchor paper (P1)	Paper P2	Paper P3
Citation	authors, title, venue, year		
Input	image, text, signal, table, graph		
Output	class, value, labeled pixels, text		
Metric	e.g. AUROC, Dice, F1		
ML task	task name		

Anchor paper worksheet (2 of 2)

4 Name the task, find its non-biomedical counterpart one paper from CVPR / NeurIPS / ICML / ACL ... with the same task on non-biomedical data (arXiv is fine here)

Task name <i>same task in both domains</i>	Non-biomedical paper <i>authors, title, venue, year</i>
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5 Compare the three papers one phrase per cell; from the abstract, Fig. 1 and the first paragraph of the data section

	P1 (anchor)	P2	P3
Motivation	<i>why they did it</i>		
Data	<i>source, size, label source</i>		
Results	<i>number + baseline</i>		
Implication	<i>clinical / scientific</i>		

6 Summarize your pitch

a. Our ML task is called *[task name]*, we apply it to *[INPUT]* to get *[OUTPUT]* in order to *[GOAL]*.

b. We propose adapting *[NON-BIOMEDICAL PAPER from Step 4]*, because *[WHY TRANSLATABLE: same input / output / metric / scale]* but *[PITFALLS: pick ≥2]*:

<i>pitfall 1</i>	<i>pitfall 2</i>
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Navigate
the
literature

What has been
done

(do I understand
others
correctly?)

Find
Research
Gap

What's still
to be done
(do I identify
worthwhile
research gap?)

Design
Research
Plan

How will I
conduct it
(did I plan
correctly?)

Conduct
Research

Research
Output
(did I do it
properly?)

Discuss
Result
Implication

What does our
result mean
(did I context-
ualize it well?)

Doing Research: the
old way

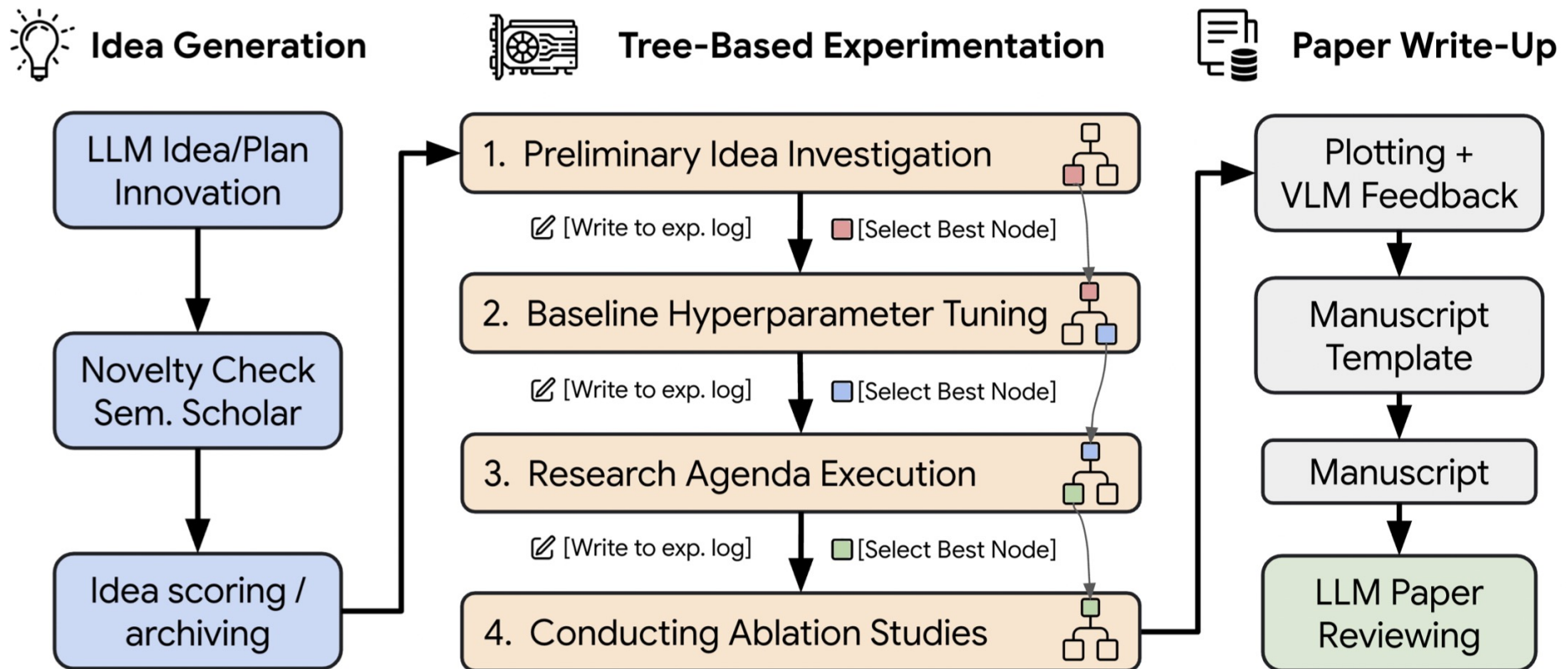
<https://github.com/changlabntu/ML2026>

The LLM way

THE AI SCIENTIST-v2: Workshop-Level Automated Scientific Discovery via Agentic Tree Search

Yutaro Yamada^{1,*}, Robert Tjarko Lange^{1,*}, Cong Lu^{1,2,3,*}, Shengran Hu^{1,2,3}, Chris Lu⁴, Jakob Foerster⁴, Jeff Clune^{2,3,5,†} and David Ha^{1,†}

^{*}Equal Contribution, ¹Sakana AI, ²University of British Columbia, ³Vector Institute, ⁴FLAIR, University of Oxford, ⁵Canada CIFAR AI Chair, [†]Equal Advising



(do I understand others correctly?)

LLM Idea/Plan Innovation

Novelty Check Sem. Scholar

Idea scoring / archiving

(do I identify worthwhile research gap?)

(did I plan correctly?)

1. Preliminary Idea Investigation

✍ [Write to exp. log]

📁 [Select Best Node]

2. Baseline Hyperparameter Tuning

✍ [Write to exp. log]

📁 [Select Best Node]

3. Research Agenda Execution

✍ [Write to exp. log]

📁 [Select Best Node]

4. Conducting Ablation Studies

(did I do it properly?)

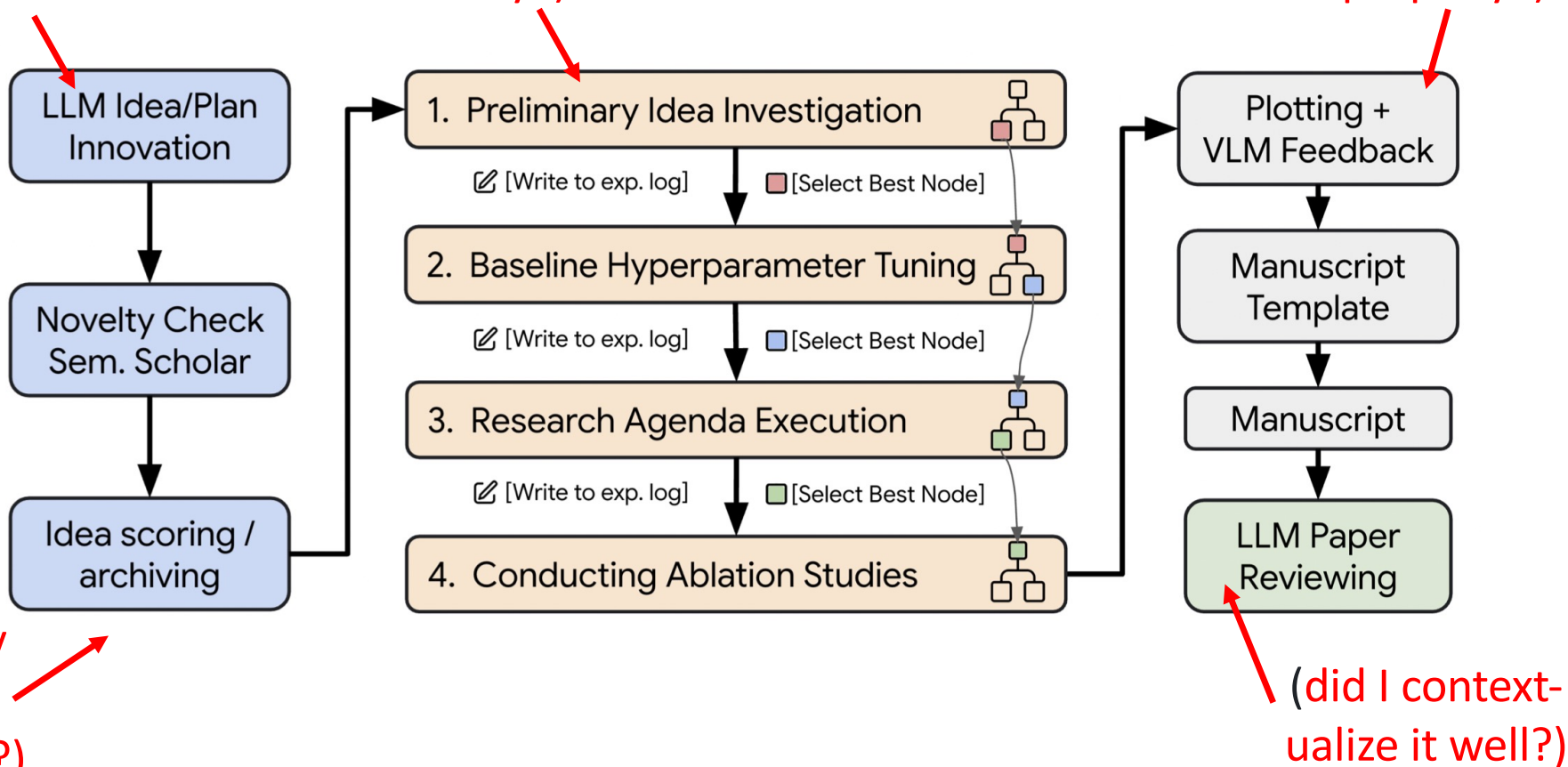
Plotting + VLM Feedback

Manuscript Template

Manuscript

LLM Paper Reviewing

(did I contextualize it well?)



Using AI “well” (in my humble opinion)

Understand how can a **task** be
breakdown into **pipelines**

Know when and where to use **judgments**

And how to connect these using **AI**

1 Step 1: pick one anchor paper

- ☐ Biomedical + machine learning
- ☐ Peer-reviewed, last 5 years
- ☐ Original study, not a review
- ☐ Not from [excluded venues]
- ☐ PubMed, Scholar, WoS, LLM helper

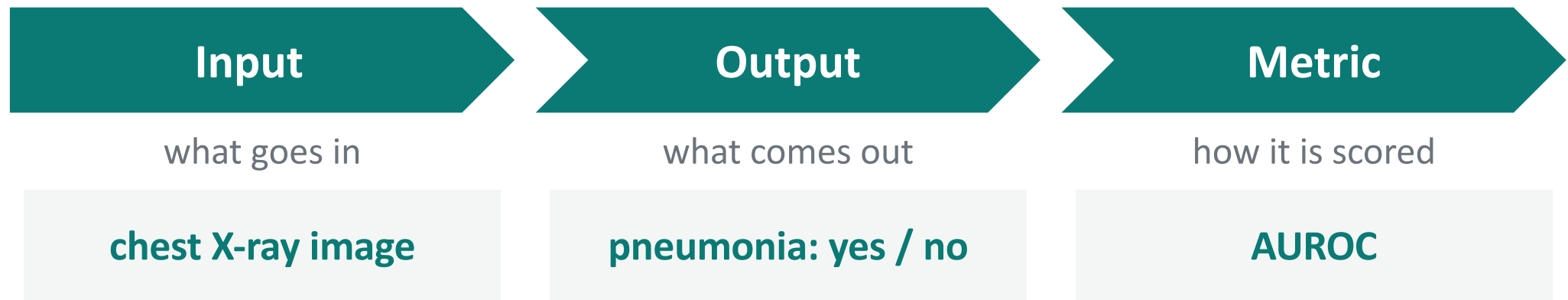
Why one?

Papers 2 and 3 must match its input and output.

Example

Pneumonia on chest X-rays, CNN, radiology journal, 2022.

2 3 Steps 2–3: state input, output and metric, then find two papers that match



Together they name the ML task: binary image classification. Ignore the model.

4 Step 4: find the same task on non-biomedical data

Biomedical (your anchor, P1)

radiology journal, 2022

Input chest X-ray

Output pneumonia yes / no

Metric AUROC

Data 112k X-rays

same task
**image
classification**

Non-biomedical counterpart

CVPR / NeurIPS / ICML / ACL, arXiv is fine

Input natural photo

Output object class

Metric top-1 accuracy

Data 1.3M photos

5 Step 5: compare motivation, data, results and implication

Row	Question it answers	Where to look	Example (anchor)
Motivation	Why? What gap?	Abstract, intro	<i>Radiologist shortage</i>
Data	Source, size, label origin	Data / methods section	<i>112k X-rays, report labels</i>
Results	Headline number vs baseline	Abstract, Fig. 1, first table	<i>AUROC 0.92 vs 0.85</i>
Implication	So what?	End of discussion	<i>Triage, not replacement</i>

6 Step 6: the pitch in two sentences

a. The problem

Our ML task is called **[task name]**, we apply it to **[INPUT]** to get **[OUTPUT]** in order to **[GOAL]**.

Image classification on chest X-rays, pneumonia yes / no, to triage urgent cases faster.

b. The proposal

We propose adapting **[NON-BIOMEDICAL PAPER from Step 4]**, because **[WHY TRANSLATABLE]** but **[PITFALLS, pick at least 2]**.

Adapt the CVPR classifier: same task, but low-contrast images and rare positives.

Pitfalls (pick two)

Domain shift

different image statistics

Label scarcity

few expert labels

Class imbalance

rare positives

Scale

thousands, not millions

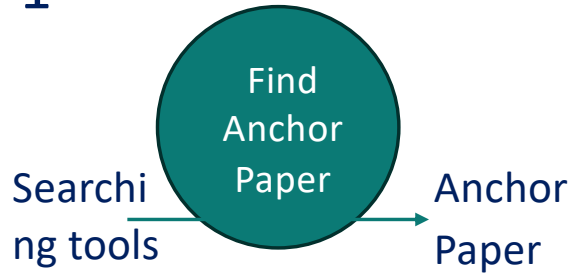
Metric mismatch

accuracy vs AUROC

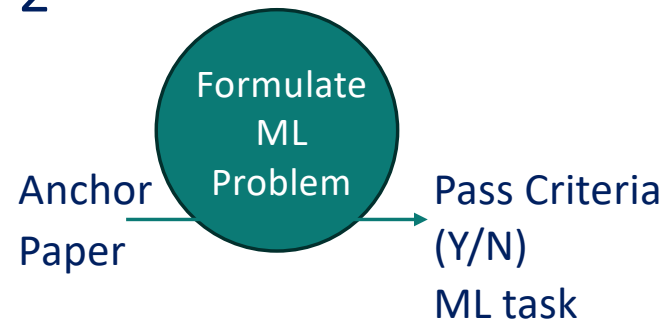
Cost of errors

a miss is not a wrong tag

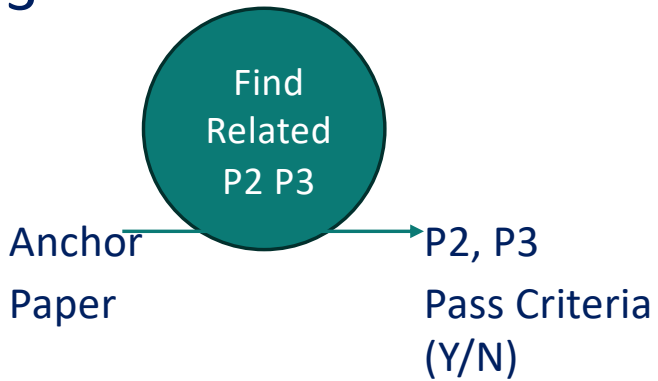
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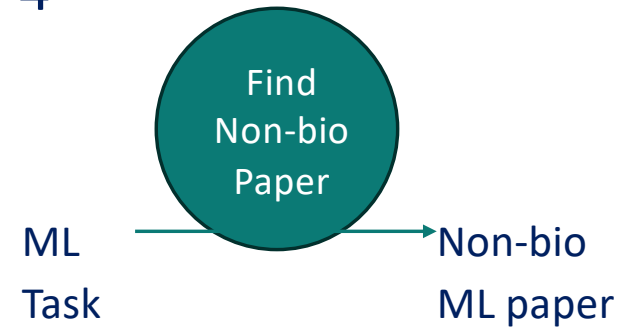
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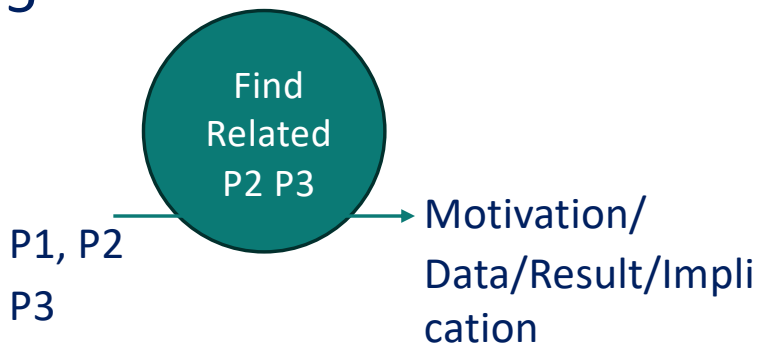
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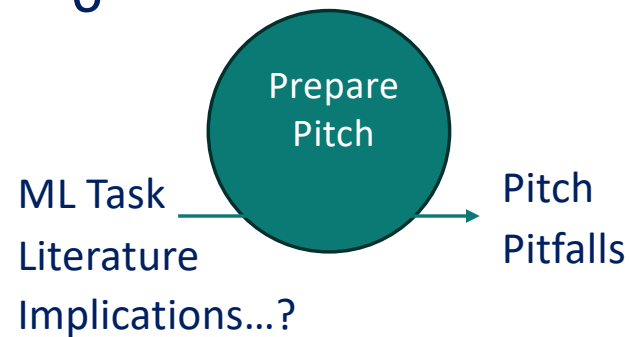
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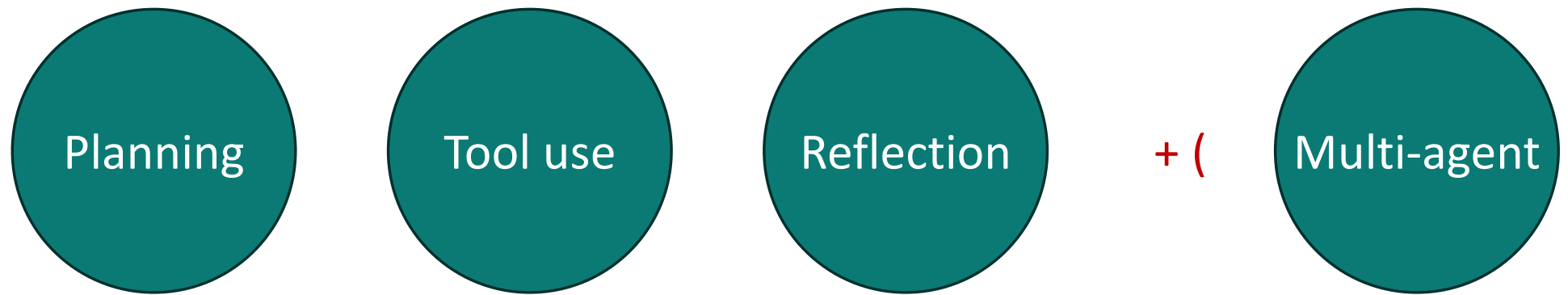


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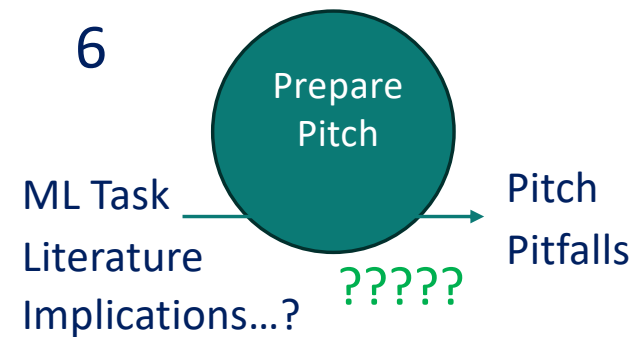
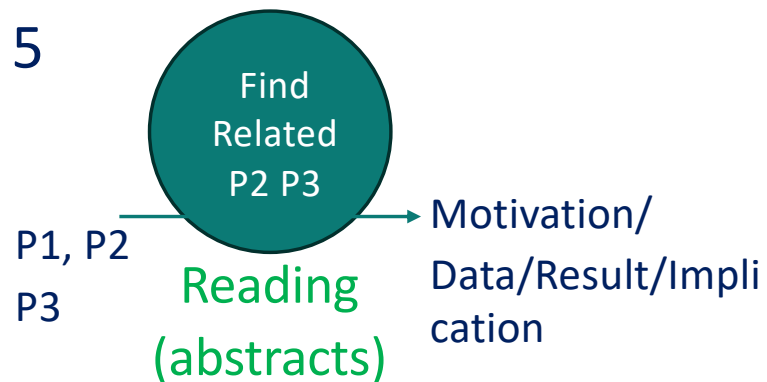
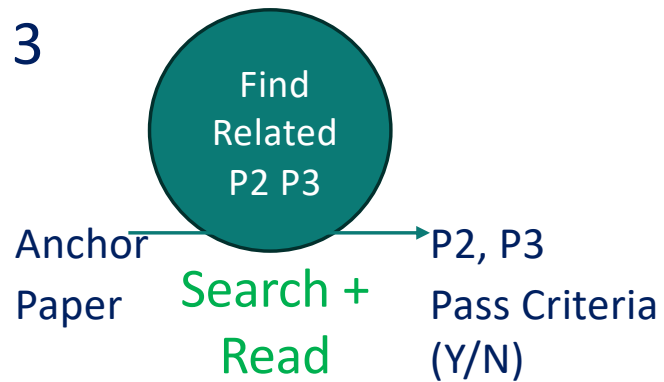
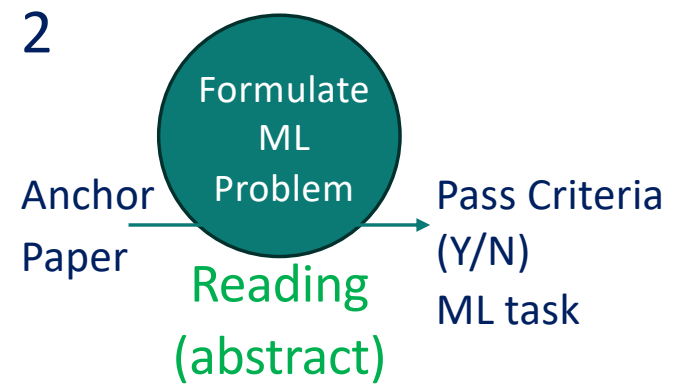
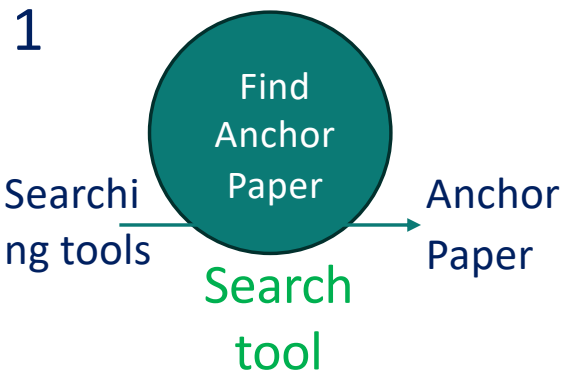


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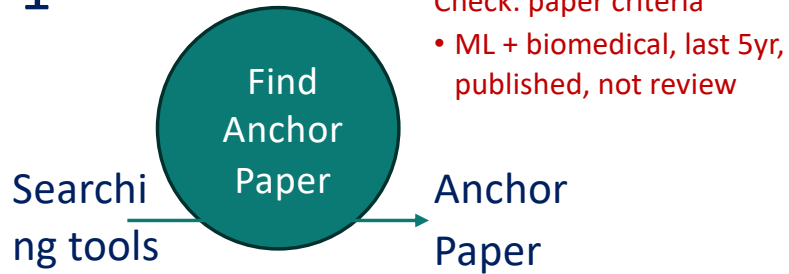




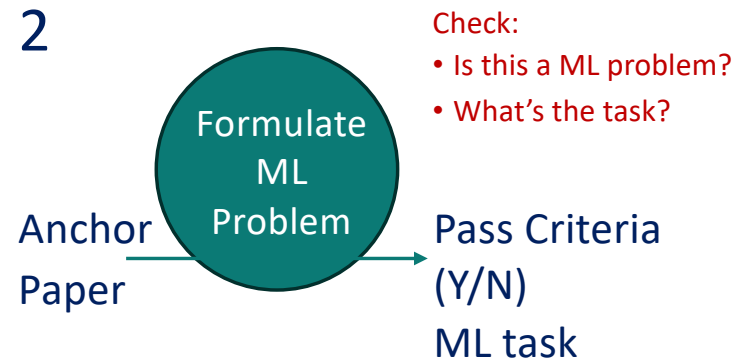
+ “Human in the loop”



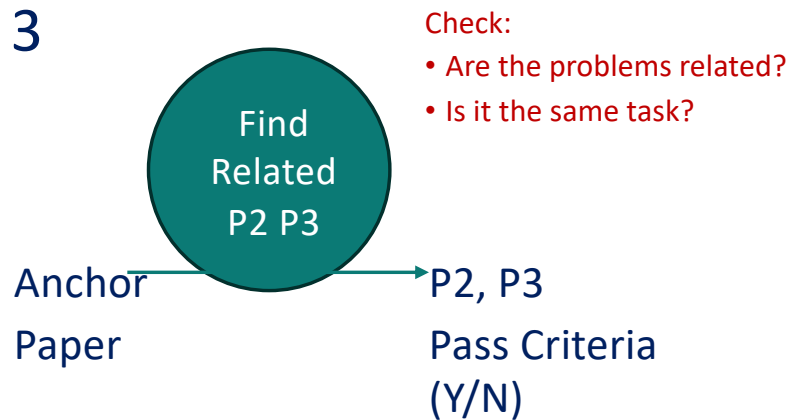
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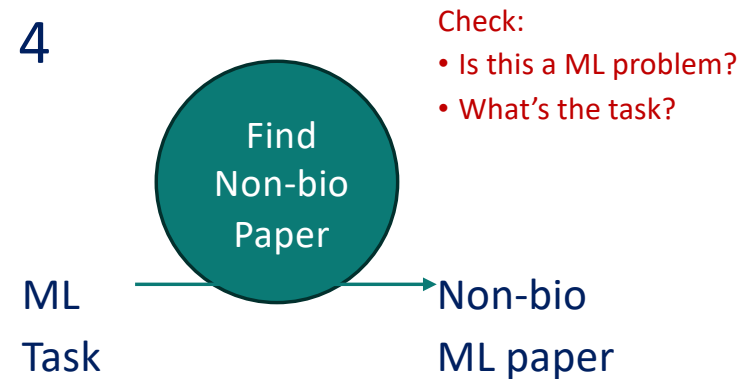
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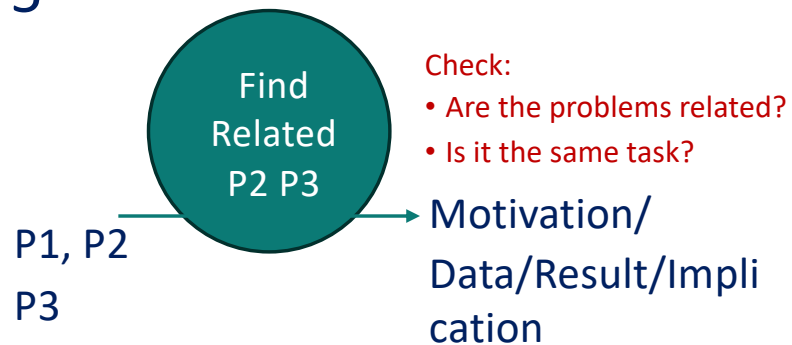
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4



5



6

